



Electrical Machines & Drives [ELEN4007/6020]

Laboratory N° 1

LEYBOLD & LABVOLT EQUIPMENT INSPECTION.

Important Safety Notes

- The following points are very important and critical for your safety when working in the laboratory. You will be working with live high voltages that can be dangerous if the correct procedures are not observed.
- All Students must wear enclosed shoes in the laboratory
- Leave your bags on the table at the entry to the laboratory so that they do not clutter your work space and no one will trip over them.
- If you are late for the laboratory you may not be allowed to attend the laboratory session as you may have missed important information relative to the experiment and safety, given by the laboratory supervisor at the beginning of the laboratory session.
- Never energise a circuit before it has been checked by your laboratory supervisor.
- Never leave unused cables and wires on your work bench.
- Never connect two different coloured wires together.
- At the completion of the laboratory, your circuit should be disconnected, wires and cables returned to their storage areas and results signed by your supervisor.
- In the case of an emergency (fire, safety, electrocution,) **Don't panic!** Immediately push the **red safety isolation button** on the wall and then immediately **dial 0000** from the phone on the wall next to the entrance of the laboratory. Your location is 206.203A.

1. OBJECTIVE

To effectively operate in a laboratory a thorough knowledge of the equipment to be used is essential. This laboratory requires the inspection each piece of Leybold equipment and the recording of the equipments performance specifications. This includes machines and control gear:

- Synchronous Machines
- Induction Motors
- Pendulum Machines
- Control Units
- Indicators and Measurement Devices

As part of this laboratory the you need to identify the types and applications of the various machines.

2. APPARATUS

LAYBOLD

Synchronous Machine:	73306
Synchronous Machine:	73307
Pendulum Machine:	73268
Induction Motor:	73281
Induction Motor:	73294
Induction Motor:	73296
Induction Motor:	73303
Inertia Load:	73314
Electrical Dynamometer:	773,2990
Control Unit:	732695
Control Unit:	72683
Machine Test Unit:	7732.901
Converter Control Unit:	773.5291
DC Power Supply	773.5295
Universal Inverter	773.5297
Rotor Starter	732.99
Frequency Converter	735.3102
Auxiliary Equipment:	72727
	73162
	72728
	732592

LABVOLT

DC Motor	8211-0A
Dynamometer	8960-15
Speed Controler	9017-00

3. PRELIMINARY:

- 3.1 Read the complete laboratory for Electrical Machines & Drives [ELEN4007], Lab1 before coming to your first laboratory session. Also check the time slot of the laboratory in which you have enrolled as most laboratory sessions are full and there will not be room for extras in most time slots.

4. LABORATORY: [20 Marks]

Each item of this first laboratory will need to be inspected and appropriate data recorded. If you are inspecting an induction motor record the type of induction motor. Take a **photograph** of each item and nameplate with your phone to include with your report.

4.1 Machine Inspection

- 4.1.1 Examine and Photograph each of the Rotating Machines listed in the Part 2 Apparatus.
- 4.1.1.1 Photograph the name plate, record all the name plate details of each machine (naming the type of Machine, number of poles and full load speed). Make sure you record which photograph represents each type of machine.
- 4.1.1.2 Identify all terminals of the machines ensuring you know the purpose of each connection and the winding arrangements.
- 4.1.1.3 Measure and record the **Resistance** of any winding associated with the machine being inspected

You will need to present all this data in a consistent tabulated form for each machine along with a photograph of each piece of equipment in the proformer on Blackboard.

4.2 Control Units

- 4.2.1 Examine and photograph the two control units listed in above in Part 2, Apparatus:
- 4.2.1.1 Record and name the types of control units and relate to the data obtained in the laboratory preliminary
- 4.2.1.2 Become familiar with each of the inputs / outputs and controls available on each of the control units as described on the Leybold brochures.

All recorded experimental results should be recorded in a logbook which you will submit for inspection at the end of semester

To answer the questions that must be included in the report you will be required to research a number of topics and concepts related to Electrical Machines.

5. REPORT TO INCLUDE:

[55 Marks]

5.1 The answers to the following questions should be included in any laboratory report discussion section.

5.1.1 The name Plate details of a 37 kW AC induction motor is shown below.

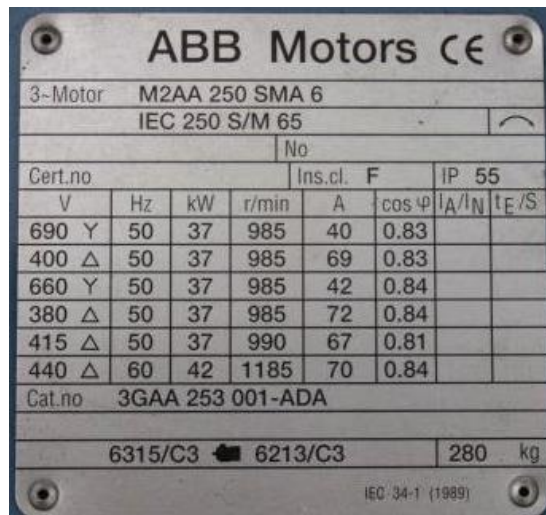


ABB Motors CE								
3-Motor M2AA 250 SMA 6								
IEC 250 S/M 65								
No								
Cert.no				Ins.cl. F		IP 55		
V	Hz	kW	r/min	A	cos φ	I _A /I _N	t _E /S	
690 Y	50	37	985	40	0.83			
400 Δ	50	37	985	69	0.83			
660 Y	50	37	985	42	0.84			
380 Δ	50	37	985	72	0.84			
415 Δ	50	37	990	67	0.81			
440 Δ	60	42	1185	70	0.84			
Cat.no 3GAA 253 001-ADA								
6315/C3						6213/C3		280 kg
IEC 34-1 (1989)								

5.1.2 Determine at 415 V the input Full Load current; [2 Marks]

5.1.3 Determine at 415 V the Full Load Power Factor; [2 Marks]

5.1.4 Explain the **IP rating of 55** and other possible IP ratings for motors; [3 Marks]

5.1.5 Tabulate the various insulation classes and in particular the **Class F** of this motor stating their temperature rating. [2 Marks]

5.1.6 With the **AID of DIAGRAMS**, explain the duty ratings of **S1, S2, S3** and **S4**, and give typical practical applications for each of the ratings. [5 Marks]

5.2.1 The Magnitude of the Induced Voltage in one phase an Alternator can be expressed as:

$$E_{\text{RMS}} = 2.22 Z f \phi \text{ Volts.}$$

Derive this expression Where

Z = N° Conductors per phase

f = Frequency

ϕ = Flux per pole

[3 Marks]

The normal expression is: $E_{\text{RMS}} = 2.22 Z k_p k_D f \Phi$ Volts.

Where k_p = Pitch Factor of winding
 k_D = Distribution Factor of winding

5.2.2 With the aid of **diagrams** derive the expression for Pitch Factor k_p .

$$k_p = \cos(\alpha/2) \quad [4 \text{ Marks}]$$

5.2.3 With the aid of **diagrams** derive the expression for Pitch Factor k_D .

$$k_D = \frac{\sin m\beta/2}{m \sin \beta/2} \quad [4 \text{ Marks}]$$

5.3.1 What is the typical generated voltage from a large three phase base load alternator (say 600 MVA)? [3 Marks]

5.3.2 Why are the armatures of large power alternators connected STAR, even when their outputs are connected directly to a Unit Transformer? [4 Marks]

5.3.3 What is a Unit Transformer? [3 Marks]

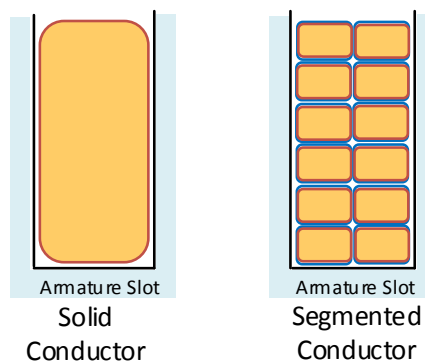
5.4. A three phase, STAR connected six (6) pole, Alternator has a rotational speed of 1000rev/min. The armature has 54 slots with eight (8) conductors per slot. The winding is double layer, short pitched by one slot, with the flux per pole $\Phi = 0.129$ Wb. Determine:

5.4.1 The No Load phase voltage. [4 Marks]

5.4.2 The No Load Line Voltage. [1 Marks]

5.5 Explain how the aging of the insulation of a machine may be tested to prevent insulation failure. [3 Marks]

5.6 The conductors of large alternators are segmented, as shown below, connected in parallel and then transposed along the length of the alternator slot. Each conductor segment is electrically insulated from the other segments.



- 5.6.1 Using electrical theory and diagrams explain why the armature conductor is segmented. **[4 Marks]**
- 5.6.2 Again using electrical theory and diagrams explain why the conductors are transposed. Include a diagram of transposed conductors. **[4 Marks]**
- 5.7.1 Calculate the maximum load current at 0.87 pf lagging, for a 790 MVA, 13.7 kV, 50 Hz, Star connected alternator. **[3 Marks]**

Tabulate all recorded laboratory data in a clear tabular form for each piece of equipment to allow easy access for future experiments.

A short form report in an on a Thumb Drive must be handed to the laboratory supervisor for marking at your next laboratory session in two weeks .

Only one report needs to be submitted per laboratory group, but each member of the group should have a copy of the final report on their own Thumb Drive.

(If you do NOT attach the Laboratory sheet to your report you will need to write a Section on Laboratory Method)

Use of the Proformer for a short form report (for Laboratory 1) may be found on Blackboard at Electrical Machines & Drives ELEN4007 . **[25 Marks]**

Late Laboratory reports will be penalised in marks. Any reports over two weeks late will not be awarded any marks (Zero).

Note: Of the 15 Marks for the Performance of the Laboratory, 5 Marks are for the participation and 10 marks for the orderly presentation of the recorded data.

End of Experiment

STUDENT NAME: _____

STUDENT NUMBER: _____

DATE (Semester 2, 2026): _____

SUPERVISOR'S SIGNATURE: _____

FOLDER: CURTIN

SubFOLDER: ELECTRICAL MACHINES & DRIVES

FILE: LAB1-ELEN4007_2026V2-6.DOC

AUTH: N.D.KAYE

DATE: JULY 2026

FOR SEMESTER 2

REPORT CHECK POINTS**TICK BOXES when Complete**As per example; **General Guidelines on Laboratory, Note-keeping and Report Writing.**

- ☐ Enclosed in Plastic Sleeve.
- ☐ PRINT **One sided only**.

Cover Sheet

- ☐ Subject Name.
- ☐ Report Title.
- ☐ Student Name.
- ☐ Time and Date Performed **Supervisors Name.**
- ☐ Date Submitted.
- ☐ Other members of Laboratory Group.
- ☐ Declaration.

Title and Summary page.(New Page)

- ☐ Clear large Font Title of Laboratory.
- ☐ Aim or Purpose without Method (approx 6 – 8 lines **Max**).
- ☐ Summary of report, was **Aim** achieved and within **what max** percentage % error.

Laboratory Tabulated Results. (New Page)

- ☐ Separated Page with **own tables**.
- ☐ Clear comparison between theoretical Pre-laboratory Results and Measured Results.
- ☐ Each Section or Table clearly titled, (**Do not used section numbers from lab sheet**).

Graphs (These are results)

- ☐ Full A4 Size graphs.
- ☐ Title on Graph.
- ☐ All Plotted points shown.
- ☐ Curve of best fit for plotted points shown.
- ☐ Both Horizontal and Vertical grid lines on Graphs.
- ☐ Comparison Curves on same graph of **Theoretical and Measured** results.

Body of Report, Discussion. (New Page)

- ☐ Interpretation of results.
- ☐ Applications of Principles Presented.
- ☐ Answer to Questions from laboratory sheet, in Paragraph Form.
- ☐ Referencing to Appendix for any derivations and **calculations using measured values**.
- ☐ Reasons for variations between Theoretical and Measured results.
- ☐ Discussion a **maximum** of two pages.

Conclusions.

- ☐ Deductions and inferences.
- ☐ Significant points.
- ☐ Was the Laboratory successful?

Appendix (New Page)

- ☐ Required derivations.
- ☐ **One** example of **each type** of calculation **using measured values**.

References

- ☐ Books used in research on report (if none used state **state lecture notes**).
- ☐ Papers etc used in construction of report.

General.

- ☐ Laboratory sheet included in report (no need to include method in if attached)
- ☐ **Photocopy** of the signed results pasted in A4 log book.
- ☐ **All report in Third Person Past Tense style.** [20 Marks]